

Exercise 1 Largest inscribed ball

Find the largest ball inside a polyhedron $\mathcal{P} = \{x \in \mathbb{R}^2 \mid 0 \leq x_1 \leq 2, 0 \leq x_2 \leq 1\}$.

Hint: Use LP description of the problem:

$$\begin{aligned} & \max_{x,r} r \\ & \text{subj. to } a_i^T x + r \|a_i\|_2 \leq b_i, \quad i = 1, \dots, m \end{aligned}$$

where r is the radius of the ball, x is vector the of center coordinates, and $a_i^T x \leq b_i$ are hyperplanes which define the polyhedron.

- 1) The problem is two dimensional and thus can be easily depicted. Draw $\mathcal{P}$ and guess which constraints will be active. Does the problem have a unique optimizer?
- 2) Solve the same problem for a triangle: $\mathcal{P} = \{x \in \mathbb{R}^2 \mid x_1 \geq 0, x_2 \geq 0, 2x_1 + x_2 \leq 2\}$. Notice that only a minor change is necessary in the previous LP formulation, while solving the problem by hand is not so trivial anymore. Solve the problem with MATLAB.

Exercise 2 Minkowski sum of invariant sets

Let X_1 and X_2 be invariant sets for the autonomous system $x(k+1) = Ax(k)$. Show that $\alpha X_1 \oplus (1-\alpha)X_2$ is also an invariant set for any $\alpha \in [0, 1]$.

Hint: The definition of the Minkowski sum of two sets X and Y is

$$X \oplus Y := \{x + y \mid x \in X, y \in Y\}.$$

Exercise 3 Invariance and feedback laws

Consider the system,

$$x(k+1) = \begin{bmatrix} 0.5 & -0.25 \\ 0.25 & 0.5 \end{bmatrix} x(k) + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} u(k),$$

with inputs constraints of the form

$$\mathcal{U} = \{u = (u^{(1)}, u^{(2)}) \mid -5 \leq u^{(i)} \leq 5, \quad i = 1, 2\}.$$

The goal of this exercise is to design a state feedback controller $u = Fx$, such that the set,

$$\mathcal{X}_F = \{x = (x^{(1)}, x^{(2)}) \mid -10 \leq x^{(i)} \leq 10, \quad i = 1, 2\},$$

is an invariant set of the closed loop system under the given actuator constraints. Compute all values of F , which satisfy the given constraints.

Hint 1: You may restrict your attention to diagonal feedback matrix of the form,

$$F = \begin{bmatrix} f_1 & 0 \\ 0 & f_2 \end{bmatrix}.$$

Hint 2: Argue why it is sufficient to only check the corner points of the set $\mathcal{X}_F$.

Exercise 4 **Maximal Invariant Set**

Consider the one-dimensional linear system

$$x(k+1) = -0.5x(k)$$

subject to the constraints

$$\mathcal{X} := \{x \mid -0.5 \leq x \leq 2\}$$

- 1) Give an algorithm for computing the maximal invariant set.
- 2) Compute the maximal invariant set for this system without MATLAB.